

Research Paper on EV Adoption and Policy Incentives

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1 Introduction

• EV penetration in the automobile industry Electric vehicles (EVs) have emerged as a pivotal technology in the global transition towards sustainable transportation. Their adoption is closely linked to environmental policy frameworks, which aim to mitigate greenhouse gas emissions and reduce urban air pollution. Governments worldwide have implemented a range of incentives to accelerate EV uptake, including direct purchase subsidies, tax rebates, and exemptions from registration fees. In addition, indirect incentives such as investments in charging infrastructure, preferential access to urban zones, and public awareness campaigns have played a significant role in shaping consumer behavior. The effectiveness of these policies depends on local market conditions, technological readiness, and the broader regulatory environment. As EV technology continues to evolve, policy incentives remain a critical lever for driving adoption and achieving environmental objectives.

- Global context of policy incentives for EV adoption To incentivize EV adoption, government policies have been introduced globally,

- Review on three streams of literature: Differentiated Market Demand estimation Indirect Network Effect EV Adoption and Environmental Policy

This study primarily contributes to the following three strands of literature. First, this study closely relates to the mature literature on differentiated product demand estimation, in which the automobile market has been heavily studied on a variety of directions. Following the foundational works of Bresnahan (1987); Berry, Levinsohn, and Pakes (1995); and Nevo (2000), this paper adopts the structural modelling feature of allowing flexible substitutional patterns. This is essential in estimating heterogeneous consumer preferences in product attributes which enables counterfactual policy analyses. However, the focus on EV is relatively fresh within the literature. The structural modelling approach of discrete choice allows for a wide-range of policy counterfactuals and as an emerging technology, EV demand estimation involves new perspectives to the traditional studies on the automobile industry.

2 Industrial Background and Policy

- Development of Chinese EV market The Chinese EV market has experienced a rapid expansion period over the last decade, and various incentives in the form of both central and disaggregated policies have contributed to the process. Back in 2009, the Chinese government first initiated the experimental policy initiative of "1,000 EVs in 10 cities", aiming at introducing new energy vehicles at 10 cities per year. To initialize EV adoption, the replacement over traditional ICE cars initially took place in public transports, taxi, and other government-led fields. However, EV penetration was unsuccessful during earlier stages due to immature industry of EV production, technologies and infrastructures. Within the same period, monetary incentives were also introduced on both supply and demand side in the form of cost rebates and purchase subsidies. In 2014, a uniform tax rebate policy was introduced for all purchases on BEVs and PHEVs. These continued to exist until the environmental policies gradually gained public attention in major cities. Many local environmental policies had taken the form of travel and registration restrictions on traditional vehicles while leaving EV unaffected. As technologies progressed, price effects combining with usage value rise accelerated EV penetration.

In the starting period, EV share rose explosively but with a strong reliance on intense monetary incentives. The side effects of subsidy fraud also came as a side effect. As a result, the attention gradually shifted from direct to indirect policies, among which infrastructure supply was key due to the unique two-sided market feature of EV. From 2017, policy exists on direct subsidies was complemented by construction cost rebate and operational subsidies for charging stations. In the meantime, foreign capital restrictions were lifted for automobile producers, following the entry of Tesla's super-factory in Shanghai, number of brands began to rise, pushing up the intensiveness of differentiated product competition. In addition, the 'Dual Credit Policy' published in 2017 effectively ensured a minimum share for EV from the supply side, resulted in large scale entries of traditional ICE automobile players to expand into EV production.

- Central/local government incentives: purchase and infrastructure

3 Data

The data I use is compiled from a variety of sources, including a commercial source that aggregated registry data from the Chinese Association of Automobile Manufacturers (CAAM), indicative price and range data from the Ministry of Industry and Information Technology (MIIT), charging station stock data from the China Electric Vehicle Charging Infrastructure Promotion Alliance (EVCIPA), the province-level demographic data National Bureau of Statistics (NBS), and supplementary information on a wide-range of government incentives at different levels.

The primary data set is the sales data where each entry contains the number of sales for a particular model within a province in a given year in the range of 2019-2023, it also contains brand, fuel-type and two supplementary product characteristics of each model: mass and power. This data set is over all types of private cars, consisting ICE cars and EVs that can be further divided into BEVs and PHEVs. The data set was purchased from a commercial source that aggregated micro data of Compulsory Traffic Accident Liability Insurance (CTALI) maintained by the CAAM. It is common practice to use this measure as proxy for car purchases in the Chinese market, as CTALI registered cars are the ones actually in-use. The number of models appears to vary across years and the availability of some are not uniform across provinces. I define a market as the unit of observation at the model-year-province level, the data set consisting of ?? models, 31 provinces over a 5-years interval, gives me a total number of ?? observations. This sample size is significant compares to other studies due to the large geographic dimension and uniquely rich diversification in the Chinese automobile market, supported by its market base.

The second data set collected information on indicative price range published by the MITT, and I use the midpoint of which as the market price. It also contains the range of each model.

I also obtain the charging station stock at the province-year level from 2018-2023.

Lastly, I combine mean and standard deviation of demographic variables for each province-year.

4 Empirical Modelling

4.1 Demand Modelling

For modelling vehicle demand side, I follow the traditional setup of differentiated product demand estimation to use a discrete choice specification with random utility maximization. In particular, assume there are $m = 1, \dots, M$ markets that is defined as a province-year pair within my context of study. Within each of the defined market, the modelling is on individual purchase decisions of $i = 1, \dots, I_m$ number of potential consumers, made over $j = 1, \dots, J$ different vehicle models in the product space. The indirect utility of consumer i , $U(i)$ of consuming product j in market m is specified as a function of both observed and unobserved product characteristics:

$$U_{ijm} = \alpha(y_i - p_{jm}) + x_{jm}\beta^{ex} + \xi_{jm} + \bar{\epsilon}_{h(j)m}^i + (1 - \rho)\bar{\epsilon}_{ijm} \quad (1)$$

where p_{jm} denotes the net product prices that includes taxes and intermediate markups, y_i the income of consumer,

5 Results

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6 Policy Counterfactuals

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7 Conclusion

8 Literature Review